

Spectral relations between oriented graphs and corresponding signed graphs

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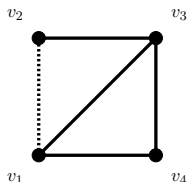
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Joint work with Zoran Stanić (University of Belgrade, Serbia)
and Francesco Belardo (University of Naples Federico II, Italy).

Some basics

Signed graphs and adjacency matrices

For a finite simple undirected graph $G = (V, E)$, a signed graph $\dot{G}$ is a pair (G, σ) where σ is the edge signature satisfying $\sigma(v_i v_j) \in \{1, -1\}$ for every $v_i v_j \in E$. In general the underlying graph G has no restrictions, however we shall be considering finite simple (no loops or multiple edges) graphs.



$$A_{\dot{G}} = \begin{pmatrix} 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

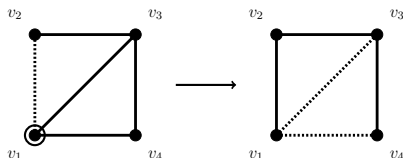
The adjacency matrix of a signed graph $\dot{G} = (G, \sigma)$ on n vertices is a square matrix of order n that is denoted by $A_{\dot{G}} = (a_{ij})_{n \times n}$ and defined by

$$a_{ij} = \begin{cases} \sigma(v_i v_j), & \text{if } v_i v_j \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Switching

A key aspect in signed graph theory is the idea of switching. This means reversing the signs of all edges between a vertex subset and its complement.

Shown below is a switching on v_1 .

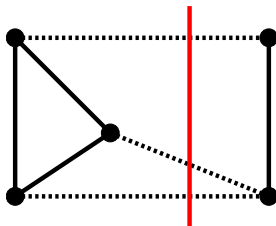


Two signed graphs $\dot{G}$ and $\dot{G}'$ on the same vertex set are switching equivalent if and only if their adjacency matrices satisfy $A_{\dot{G}} = SA_{\dot{G}'}S^{-1}$ for some diagonal matrix S which has entries of either $+1$ or -1 .

Since there is no change made to the spectrum of a matrix after it has been conjugated, any two switching equivalent signed graphs have exactly the same characteristic polynomials and eigenvalues.

Balance in signed graphs

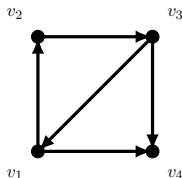
A signed graph is said to be balanced if the product of edge signs around every cycle is positive. It is well known that a signed graph is balanced if and only if it is switching equivalent to the all-positive signed graph.



The first characterisation of balance came from Harary in 1953. He said that a signed graph is balanced if and only if its vertex set can be divided into two subsets (either of which may be empty), so that each edge between the subsets is negative, and each edge within a subset is positive.

Oriented graphs and skew-adjacency matrices

For a finite simple undirected graph $G = (V, E)$, an oriented graph G' is a pair (G, ω) where ω is the edge orientation satisfying $\omega(v_i v_j) \in \{v_i, v_j\}$ for every $v_i v_j \in E$. The terms oriented edge and arc are used interchangeably.



$$S_{G'} = \begin{pmatrix} 0 & 1 & -1 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 \end{pmatrix}$$

The skew-adjacency matrix of an oriented graph $G' = (G, \omega)$ on n vertices is a square matrix of order n that is denoted by $S_{G'} = (s_{ij})_{n \times n}$ and defined by

$$s_{ij} = \begin{cases} 1, & \text{if } \omega(v_i v_j) = v_j, \\ -1, & \text{if } \omega(v_i v_j) = v_i, \\ 0, & \text{otherwise.} \end{cases}$$

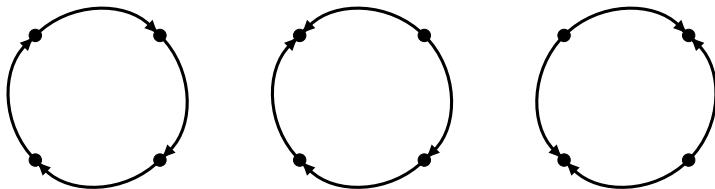
Signed and oriented spectra

Since the adjacency matrix of a signed graph is symmetric, all of its eigenvalues are real. Observe that the spectrum of the negation $-\dot{G}$ is the negation of the spectrum of $\dot{G}$.

Since the skew-adjacency matrix of an oriented graph is skew-symmetric, the skew-spectrum of G' consists of purely imaginary numbers, and the nonzero eigenvalues occur in complex conjugate pairs.

As a consequence, the skew-spectrum of G' is always symmetric with respect to the origin, and so we are going to be most concerned with signed graphs that have a symmetric spectrum.

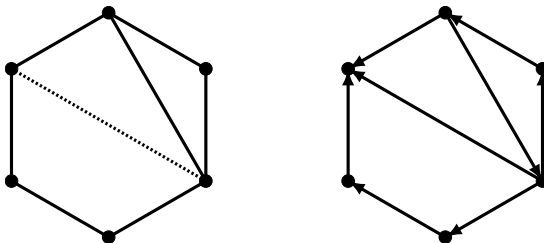
Uniformity in oriented graphs



We say that an oriented even cycle C'_{2m} is oriented uniformly if, when traversing the cycle, one passes through an even (respectively odd) number of edges oriented in the route direction when m is even (odd).

A canonical orientation of a bipartite graph is one in which all edges are oriented from one colour class to the other. In such an orientation, every cycle is oriented uniformly.

Spectral similarity



Let $\dot{G} = (G, \sigma)$ be a signed graph on n vertices with spectrum $\text{spec}(\dot{G}) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$.

Then we say that an oriented graph $G' = (G, \omega)$ on n vertices is spectrally similar to $\dot{G}$ if G' has skew-spectrum $\text{spec}(G') = \{i\lambda_1, i\lambda_2, \dots, i\lambda_n\}$, or, when

$$\text{ispec}(\dot{G}) = \text{spec}(G').$$

Preliminary results

An association

If $S_{G'}$ is the skew-adjacency matrix of an oriented graph $G' = (G, \omega)$ and $A_{\dot{G}}$ is the adjacency matrix of a signed graph $\dot{G} = (G, \sigma)$, then $-S_{G'}^2 = A_{\dot{G}}^2$ holds whenever

$$-\sum_{k=1}^n s_{ik}s_{kj} = \sum_{k=1}^n a_{ik}a_{kj}, \quad \text{for } 1 \leq i \leq j \leq n.$$

We say that the signature σ is associated with the orientation ω , and also that $\dot{G}$ is associated with G' , if

$$\sigma(ik)\sigma(jk) = \omega(ik)\omega(jk)$$

holds for every pair of edges ik and jk . Clearly, being associated is a symmetric relation.

An association

We have the following useful results.

Lemma

If σ is associated with ω , then $A_G^2 = -S_{G'}^2$.

Theorem

For a graph G and an orientation ω , there exists a signature σ associated with ω if and only if G is bipartite.

The spectral relation $\text{ispec}(\dot{G}) = \text{spec}(G')$ is a direct consequence of the previous two statements.

Spectral relations

If G' is an oriented graph, then its bipartite double $\text{bd}(G')$ is the oriented graph whose skew-adjacency matrix is the Kronecker product $S_{\text{bd}(G')} = S_{G'} \otimes A_{K_2}$, where K_2 is the complete graph with two vertices.

While previously the bipartite double was used to establish spectral relations between signed and oriented graphs with non-bipartite underlying graphs, we aim to provide more findings about spectrally similar pairs which remain in their non-bipartite forms.

We have the following lemma.

Lemma

Let $\dot{G} = (G, \sigma)$ be a signed graph and let $G' = (G, \omega)$ be an oriented graph defined on the same underlying graph of order n . Suppose that the spectrum of $\dot{G}$ is symmetric with respect to the origin. If $A_{\dot{G}}^2 = -S_{G'}^2$, then $\dot{G}$ and G' are spectrally similar.

Signed coefficient theorem

The coefficient theorem for the characteristic polynomial of a signed graph is as follows.

Theorem

Let $\dot{G} = (G, \sigma)$ and $\phi(\dot{G}, \lambda) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n$ be a signed graph and its characteristic polynomial, respectively. Then

$$a_i = \sum_{B \in \mathcal{B}_i} (-1)^{p(B)} 2^{|C(B)|} \sigma(B)$$

where $\mathcal{B}_i$ is the set of basic figures on i vertices in G , $p(B)$ is the number of connected components in B , $C(B)$ is the set of cycles in B , and $\sigma(B) = \prod_{C \in C(B)} \sigma(C)$ with $\sigma(B) = 1$ when $|C(B)| = 0$.

Oriented coefficient theorem

The analogous coefficient theorem for the skew-characteristic polynomial of an oriented graph is as follows.

Theorem

Let $G' = (G, \omega)$ and $\phi(G', \lambda) = \lambda^n + s_1\lambda^{n-1} + \cdots + s_{n-1}\lambda + s_n$ be an oriented graph and its skew-characteristic polynomial, respectively. Then $s_i = 0$ when i is odd and

$$s_i = \sum_{U \in \mathcal{U}_i^e} (-1)^{p(U)} 2^{|C(U)|} \omega(U)$$

when i is even, where $\mathcal{U}_i^e$ is the set of basic figures on i vertices in G containing no odd cycles, $p(U)$ is the number of connected components in U , $C(U)$ is the set of cycles in U , and $\omega(U) = \prod_{v_i v_j \in E(U)} \omega(v_i v_j)$.

Characteristic polynomials

Let $\dot{G}$ be a signed graph with characteristic polynomial

$$\phi(\dot{G}, \lambda) = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_{n-1} \lambda + a_n$$

and let G' be an oriented graph with skew-characteristic polynomial

$$\phi(G', \lambda) = \lambda^n + s_1 \lambda^{n-1} + \cdots + s_{n-1} \lambda + s_n.$$

For spectral symmetry we require a_i to be equal to zero for all odd values of i , a property which is held by each s_i with odd i .

If $\dot{G}$ and G' are to be spectrally similar then we require $s_i = |a_i|$ for all i . As a matter of fact, we have the requirement $a_{2i} = (-1)^i s_{2i}$.

Comparing cycles

We have the following lemma.

Lemma

Let $A = \begin{pmatrix} O & X \\ X^T & O \end{pmatrix}$ and $S = \begin{pmatrix} O & X \\ -X^T & O \end{pmatrix}$ be two real matrices. Then $\text{ispec}(A) = \text{spec}(S)$.

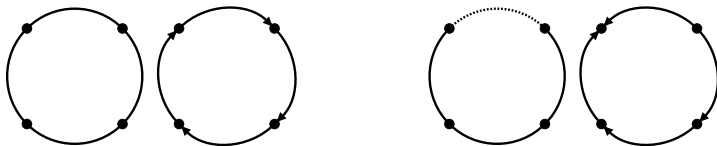
From this we can see that for a bipartite graph G , the balanced signed graph $\dot{G} = (G, \sigma)$ is spectrally similar to the canonically oriented graph $G' = (G, \omega)$.

This same idea is represented in the following theorem.

Theorem

Let G be a bipartite graph and let ω be an orientation of G . Then $G' = (G, \omega)$ is spectrally similar to $\dot{G} = (G, +)$ if and only if every even cycle is oriented uniformly in G' .

Comparing cycles and basic figures



Combining the previous lemma and theorem allows us to describe exactly when the signatures and orientations of even cycles give spectral similarity.

Lemma

Let $\dot{C}_{2m} = (C_{2m}, \sigma)$ be a signed cycle of length $2m$. We have that

- (a) when $\dot{C}_{2m}$ is balanced it is spectrally similar to a uniformly oriented cycle C'_{2m} ,
- (b) when $\dot{C}_{2m}$ is unbalanced it is spectrally similar to a non-uniformly oriented cycle C'_{2m} .

Other tools

Without giving the full details, here are some more tools we make use of.

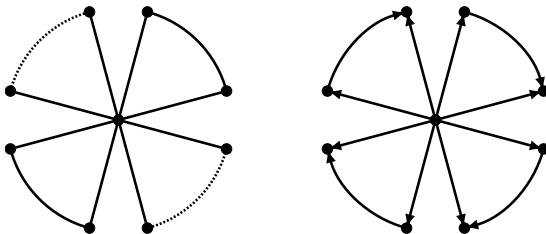
- Schwenk-like formulas from F. Belardo, E.M. Li Marzi, S. K. Simić.
- Schur's complement formula.
- The coronal of an $n \times n$ matrix M with all row sums equal to k ,

$$\chi_M(\lambda) = \mathbf{1}_n^T (\lambda I_n - M)^{-1} \mathbf{1}_n = \frac{n}{\lambda - k}.$$

- Sign-symmetry: a signed graph $\dot{G} = (G, \sigma)$ is said to be sign-symmetric if $\dot{G}$ is switching isomorphic to its negation $-\dot{G} = (G, -\sigma)$. This implies that $\dot{G}$ is spectrally symmetric.
- When the set $\mathcal{B}_i$ of basic figures coincides with the set $\mathcal{U}_i^e$ of basic figures containing no odd cycles, and each balanced (unbalanced) even cycle in the signed graph is oriented uniformly (non-uniformly) in the oriented graph, $\dot{G}$ and G' are spectrally similar.

Infinite families of graphs

Friendship graphs

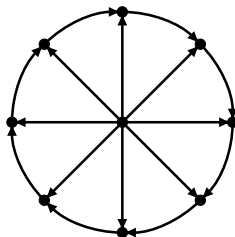
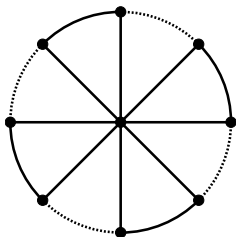


Signed: $\text{Friend}_{a,b}^\sigma$ constructed by joining a copies of the balanced cycle graph $(C_3, +)$ and b copies of the unbalanced cycle graph $(C_3, -)$ with a common, central vertex.

Oriented: $\text{Friend}_{a,b}^\omega$, where all edges incident with the central vertex are oriented from the centre to the outer vertex, and the remaining edges are oriented in any direction.

Proof: Schwenk-like formulas $\rightarrow$ select central vertex to ‘delete’ $\rightarrow$ left with paths to deal with $\rightarrow$ clearly cospectral.

Cone graphs

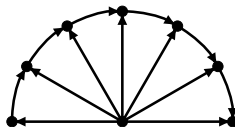
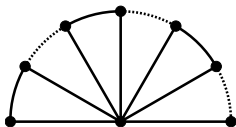


Signed: $\text{Cone}_{m,2n}^\sigma$ constructed by first making the join of C_{2n} and $\overline{K}_m$, then taking the edges from C_{2n} to be alternatingly positive and negative, and all of the edges between $\overline{K}_m$ and C_{2n} to be positive.

Oriented: $\text{Cone}_{m,2n}^\omega$ constructed on the same underlying graph, where the edges of C_{2n} are oriented in a common direction around the cycle and the remaining edges are oriented from the vertices in $\overline{K}_m$ to the vertices in C_{2n} .

Proof: Block forms $\rightarrow$ Schur's complement $\rightarrow$ coronal lemma $\rightarrow$ compare polynomials.

Fan graphs

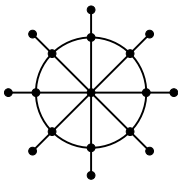


Signed: $\text{Fan}_{m,n}^\sigma$ for odd n constructed by first making the join of P_n and $\overline{K}_m$, then taking the edges from P_n to be alternatingly positive and negative, and all of the edges between $\overline{K}_m$ and P_n to be positive.

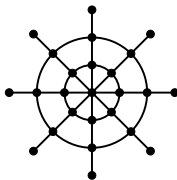
Oriented: $\text{Fan}_{m,n}^\omega$ constructed on the same underlying graph, where the edges of P_n are oriented in a common direction along the path and the remaining edges are oriented from the vertices in $\overline{K}_m$ to the vertices in P_n .

Proof: Sign-symmetry $\rightarrow$ graph is spectrally symmetric $\rightarrow$ show that $A^2 = -S^2$.

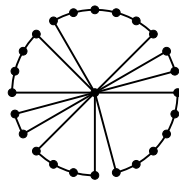
Some more families



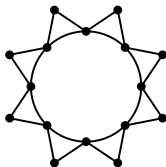
Helm



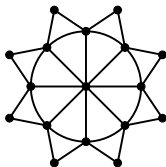
Web



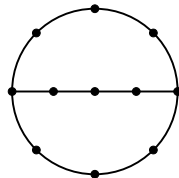
Bouquet



Sun1



Sun2



Theta

Unicyclic and bicyclic graphs

Unicyclic graphs

We have the following result from Li and Huang which characterises all unicyclic signed graphs with symmetric spectrum.

Corollary

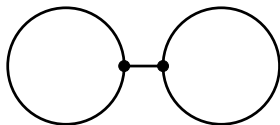
Let G be a unicyclic graph and let $\dot{G} = (G, \sigma)$, then $\dot{G}$ has a symmetric spectrum if and only if G has an even cycle.

Using this and the work done previously by Stanić, we obtain the following relation.

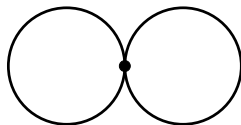
Proposition

For any unicyclic signed graph with symmetric spectrum there exists an oriented graph with a similar skew-spectrum.

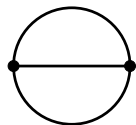
Bicyclic graphs



Type I



Type II



Type III

Let B be a bicyclic graph. There are three distinct ways in which the cycles can appear.

If they share no vertices, we call this a bicyclic graph of Type I.

If they have exactly one common vertex, we call this a bicyclic graph of Type II.

If they have more than one common vertex, we call this a bicyclic graph of Type III.

Bicyclic graphs

We have another result from Li and Huang which, in this instance, characterises all bicyclic signed graphs with symmetric spectrum.

Corollary

Let G be a bicyclic graph and let $\dot{G} = (G, \sigma)$, then $\dot{G}$ has a symmetric spectrum if and only if one of the following holds

- (a) both cycles of G are even,*
- (b) there are two odd cycles C_1 and C_2 such that $M(G - C_1, \lambda) = M(G - C_2, \lambda)$ and $\sigma(C_1)\sigma(C_2) = -1$.*

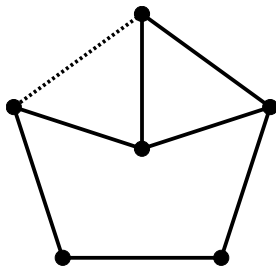
We proceed with the following result.

Proposition

For any bicyclic signed graph of Type II or Type III with symmetric spectrum there exists an oriented graph with similar skew-spectrum.

Further observations

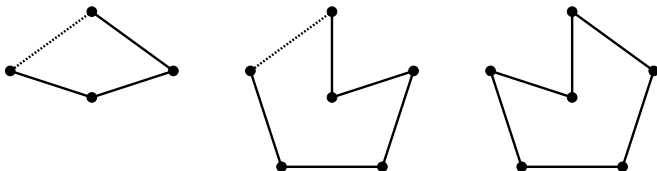
Tricyclic graphs



We began with a spectrally symmetric tricyclic signed graph that is not sign-symmetric alongside a spectrally similar oriented graph.

Here, we have a tricyclic signed graph which is sign-symmetric, and so too has a symmetric spectrum, but for which no corresponding oriented graph exists.

Tricyclic graphs

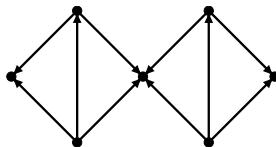
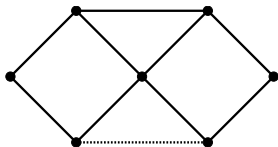
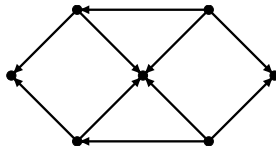
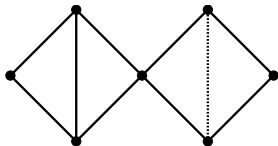


When attempting to suitably orient the edges of this graph, the problem seems to be in the way that the edges present in the 4-cycle and the two 6-cycles overlap. It is impossible to fix the issue when using this underlying graph and the lemma regarding basic figures cannot be employed.

Identifying the existence of a spectrally similar oriented graph for a tricyclic or higher signed graph seems to be a more challenging problem, lacking an immediate intuitive solution.

Pairs on different underlying graphs

Here we have two pairs of graphs which are spectrally similar yet do not have the same underlying graphs.



For the top pair we have $\{2.715, 1.275, 1, 0, -1, -1.275, -2.715\}$ and for the bottom we have $\{2.645, 1.732, 0, 0, 0, -1.732, -2.645\}$.

Pairs on different underlying graphs

There appears to some relation between the pairs, possibly arising from a form of local switching in the underlying graphs. For instance, in the examples on the previous slide, edges seem to transform from vertical to horizontal and vice versa.

However, even after investigating the eigenvalues, eigenvectors, Rayleigh quotients, and so on, it is still unclear as to what is actually happening with respect to the spectral similarity of such pairs.

As with the tricyclic graphs, identifying the existence and connections of spectrally similar pairs on different underlying graphs appears to be a more difficult task.

Thank you!